

Department of Mathematics
Motilal Nehru National Institute of Technology Allahabad
Mathematics II (MAN12101)
Course Instructor: Dr. Naren Bag
Assignment 2(Unit II)

1. Define eigenvalues and eigenvectors of a matrix. Find the eigenvalues and eigenvectors for the

matrix $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$. Also find its eigenvectors.

2. State and prove the Cayley-Hamilton theorem and verify it for the matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}$

3. Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ and find its inverse. Also express

$A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$ as a linear polynomial in A .

4. If a matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$, find the matrix A^{32} , using Cayley-Hamilton theorem.

5. If $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$, evaluate A^{-1} , A^{-2} and A^{-3} .

6. Find the characteristic equation of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence compute A^{-1} . Also find the matrix representation by $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$.

7. Find the eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$. Hence, find the modal matrix P of A and diagonalize it.

8. Find the matrix P which transforms the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ to the diagonal form. Hence calculate A^4 .

9. The eigenvectors of a 3×3 matrix A corresponding to the eigenvalues 2, 2, 4 are $[-2, 1, 0]^T$, $[-1, 0, 1]^T$ and $[1, 0, 1]^T$ respectively. Find the matrix A .

10. Show that the matrix $A = \begin{bmatrix} 3 & 1 & -1 \\ -2 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$ is diagonalizable. Hence find the modal matrix P such that $P^{-1}AP$ is a diagonal matrix.

11. Examine whether the given matrix A is diagonalizable. If so, then diagonalize it. Where

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 2 & 0 \\ 1 & 2 & 3 \end{bmatrix}$$

12. Determine the nature of the quadratic forms.

(i). $Q = x_1^2 + 2x_2^2 + 3x_3^2 + 2x_2x_3 - 2x_3x_1 + 2x_1x_2$

(ii). $Q = 4x_1^2 + x_2^2 + 15x_3^2 - 4x_1x_2$

(iii). $Q = -4x_1^2 - 2x_2^2 - 13x_3^2 - 4x_1x_2 - 8x_2x_3 - 4x_3x_1$

13. Reduce the quadratic form $Q = x_1^2 + 5x_2^2 + x_3^2 + 2x_1x_2 + 2x_2x_3 + 6x_3x_1$ to a canonical form using orthogonal transformation and hence find its nature, rank, signature and index.

14. Obtain the symmetric matrix B of the quadratic form

$$Q = x_1^2 + 2x_1x_2 - 4x_1x_3 + 6x_2x_3 - 5x_2^2 + 4x_3^2.$$

15. Define canonical form of a quadratic form. Reduce the quadratic form

$$Q = 2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_2 - 2x_2x_3 - 2x_3x_1 \text{ to canonical form through an orthogonal transformation.}$$

Find the index and signature of the quadratic form Q .

16. Reduce the following quadratic form into canonical form:

$$3x_1^2 + 3x_2^2 + 3x_3^2 + 2x_1x_2 + 2x_1x_3 - 2x_2x_3. \text{ Also find the matrix of transformation to reduce the above quadratic form into canonical form.}$$

17. The matrix $A = \begin{bmatrix} a & h \\ h & b \end{bmatrix}$ is transformed to the diagonal form $D = P^{-1}DP$, where

$$P = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}. \text{ Find the value of } \theta \text{ which gives this diagonal transformation.}$$

18. Find a matrix P which transform the matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$, to diagonal form. Hence find A^6 .